

Ethan Wong

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EDUCATION

McMaster University

B. Eng, Computer Engineering

Sep. 2023 – Present

Hamilton, ON

- **GPA: 3.9/4.0**
 - Faculty of Engineering Award of Excellence (2023), Dean's Honour List (2024)
- **Coursework:** Digital Systems Design, Data Structures and Algorithms, Software Development, Electronic Devices and Circuits, Circuits and Waves, Microprocessor Systems
- Member of Solar Car Strategies Team
 - Designed and implemented a Python algorithm to collect traffic data from the Overpass API for race simulations

WORK EXPERIENCE

AMD

ASIC Design Intern

May 2026 - Aug. 2027

Markham, ON

- Researched and introduced Tessent simulation manager for large-scale parallel DFT verification flows, including ATPG fault simulation and miscompare masking for BIST fault injection
- Designed and deployed production dashboards tracking DFT metrics (fault coverage, ATPG target status, FaultSim) in TypeScript/PHP/Python; migrated legacy JavaScript codebases to TypeScript and built CI/CD pipelines via GitHub Actions deploying to Kubernetes-hosted infrastructure
- Developed internal Python tooling integrating Weaviate vector search, LLM-generated debug summaries, LDAP queries, and Confluence API automation; built an LLM chatbot with live database access for a DFT engineering dashboard, including system prompt design

Pedalheads

Instructor

Jun. 2023 - Aug. 2023

Unionville, ON

- Taught cycling safety and techniques to dozens of campers, organizing equipment at the start and end of each day and maintaining communication with parents on progress updates

PROJECTS

Room Scanner [C++/Python]

- Built a 360° LiDAR scanner on an MSP432E401Y using a VL53L1X ToF sensor, stepper motor control, I2C data acquisition, and UART streaming; processed point-cloud data in Python and visualized results with NumPy/Open3D

Image Decompressor [Verilog/Python]

- Designed an RTL image decompression pipeline on a Cyclone IV FPGA to reconstruct 192x144 RGB images – lossless bitstream decoding, inverse DCT, and chroma upsampling – achieving >85% multiplier utilization; verified against a Python golden model in ModelSim, with a 16-bit SRAM/32-bit block RAM memory interface

2D Platformer [C#/Unity]

- Designed a 2D platformer in Unity (C#) applying OOD principles to enemies and consumables

SKILLS & INTERESTS

- **Technologies:** Verilog, Python, C, C++, C#, TypeScript, JavaScript, Git, Kubernetes, HTML, CSS
- **Interests:** Hockey, Reading